

Math 241, F12
Quiz 9

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

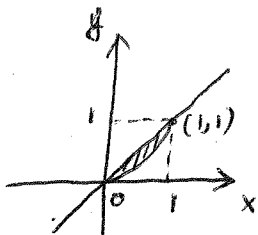
Problem 1. (6 points) Calculate the following integrals:

$$\begin{aligned} \text{(a)} \quad & \int_0^1 \int_1^2 (4x^3 - 9x^2y^2) dy dx \\ &= \int_0^1 [4x^3y - 3x^2y^3]_{y=1}^{y=2} dx \\ &= \int_0^1 4x^3 - 21x^2 dx \\ &= [x^4 - 7x^3]_{x=0}^{x=1} \\ &= -6 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad & \iint_D y^2 e^{xy} dA, \text{ with} \\ & D = \{(x, y) \mid 0 \leq y \leq 4, 0 \leq x \leq y\} \\ &= \int_0^4 \int_0^y y^2 e^{xy} dx dy \\ &= \int_0^4 [ye^{xy}]_{x=0}^{x=y} dy \\ &= \int_0^4 ye^{y^2} - y dy \\ &= \left[\frac{e^{y^2}}{2} - \frac{y^2}{2} \right]_{y=0}^{y=4} \\ &= \left(\frac{e^{16}}{2} - 8 \right) - \left(\frac{1}{2} \right) \\ &= \frac{e^{16}}{2} - \frac{17}{2} \end{aligned}$$

Problem 2. (4 points). Let D be the region enclosed by $y = x$ and $y = x^4$. Find the double integral $\iint_D x + 2y dA$.

Solution:



$$\begin{aligned} \begin{cases} y=x \\ y=x^4 \end{cases} &\Rightarrow x^4=x \Rightarrow x(x^3-1)=0 \Rightarrow x(x-1)\underbrace{(x^2+x+1)}_{>0}=0 \\ &\Rightarrow x=0 \text{ or } x=1 \\ &\quad \downarrow \quad \downarrow \\ &\quad y=0 \quad y=1 \end{aligned}$$

$$\Rightarrow D = \{(x, y) \mid 0 \leq x \leq 1, x^4 \leq y \leq x\}$$

$$\begin{aligned} \iint_D x + 2y dA &= \int_0^1 \int_{x^4}^x x + 2y dy dx = \int_0^1 [xy + y^2]_{y=x^4}^{y=x} dx \\ &= \int_0^1 2x^2 - x^5 - x^8 dx \\ &= \left[\frac{2}{3}x^3 - \frac{x^6}{6} - \frac{x^9}{9} \right]_0^1 \\ &= \frac{2}{3} - \frac{1}{6} - \frac{1}{9} = \frac{2}{3} - \frac{5}{18} = \frac{7}{18} \end{aligned}$$